

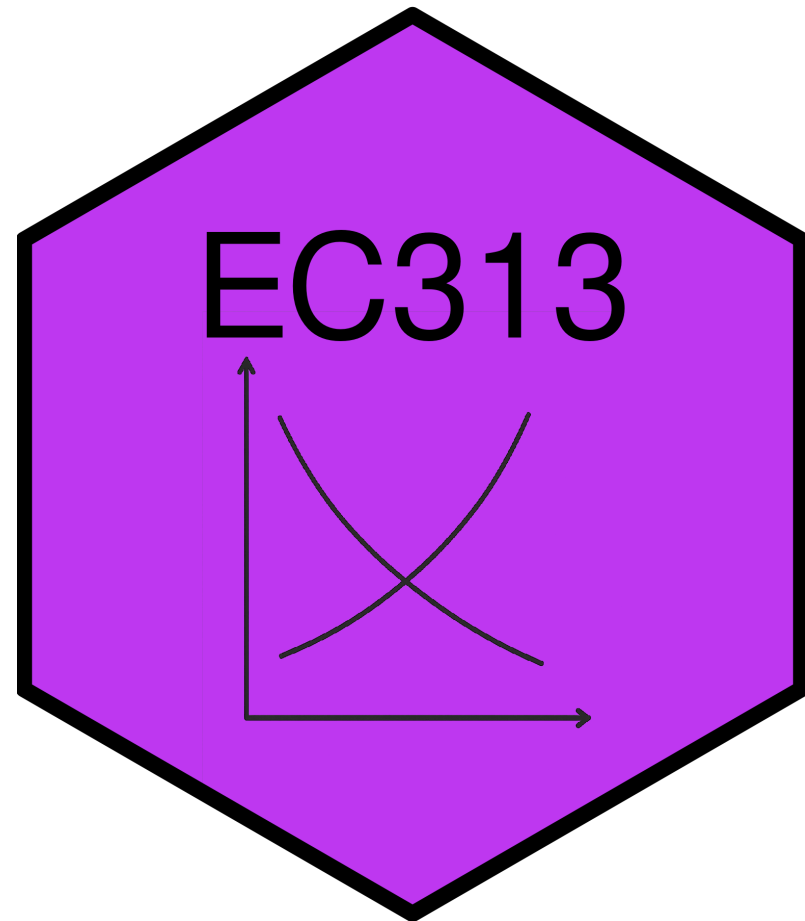
Efficient and Equitable Taxation

EC313 - Public Economics: Taxation

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Goals of This Section

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In this section we will:

- Lay out model for optimal taxation
- Derive the Ramsey Rule for optimal commodity taxation
- Discuss optimal user fees
- Discuss optimal income taxation
- Discuss political economy and time inconsistency
- Discuss other criteria for tax design

Optimal Taxation

Introduction

- As we have discussed, government raise taxes mainly to raise **revenue**
- But taxes involve costs
 - The direct cost of the revenue raised
 - Additional costs from distortions in behaviour (**excess burden**)
- Given that revenue needs to be raised, how should taxes be designed to minimize costs?
- **Optimal taxation** addresses this question

Model

- Question of optimal taxation is answered with a model
- A representative citizen consumes two goods, X and Y
 - Prices of these goods are P_X and P_Y
- Person can also work and earn wage w
 - Person has T hours available for work (h) or leisure (L)
 - These are the only two uses of time, so $T = h + L$
- Their budget constraint in this context is

$$w(T - L) = P_X X + P_Y Y$$

- Constraint says that income is split between spending on goods X and Y

Model

- If you rearrange the budget constraint, you can write it as

$$wT = P_X X + P_Y Y + wL$$

- Shows that **full income** is split between spending on goods X and Y and spending on leisure
 - Full income is wT because if all time was worked, income would be wT
 - Price of leisure is w because each hour of leisure foregoes w in earnings
- Suppose we tax goods X , Y , and L at rate t
- Then the budget constraint becomes

$$wT = (1 + t)P_X X + (1 + t)P_Y Y + (1 + t)wL$$

Model

- Rearranging gives

$$wT = (1 + t)(P_X X + P_Y Y + wL)$$

$$\frac{wT}{1 + t} = P_X X + P_Y Y + wL$$

Key Insight

A tax on all goods and leisure is equivalent to a reduction in full income by a factor of $\frac{1}{1+t}$

- We saw this in one of the practice questions

Model

! Why Taxing Everything Equally Is Non-Distortionary

If we could tax all goods and leisure at the same rate, we would **not** distort behaviour because **relative prices are unchanged**. The commodity taxes are equivalent to a **lump-sum tax** that reduces income.

- But in practice, we **cannot** tax all goods and leisure at the same rate
 - We cannot tax leisure directly
 - Can only tax X and Y in this setup
- Taxing some goods (X and Y) but not others (L) distorts relative prices and leads to **excess burden**

Ramsey Rule

- Question then becomes: If we cannot tax all goods at the same rate, how should we tax them?
- One option is **neutral taxation**: taxing X and Y at the same rate
 - This is **simple and easy to administer**
 - But it still **distorts behaviour** and creates excess burden

! The Ramsey Rule

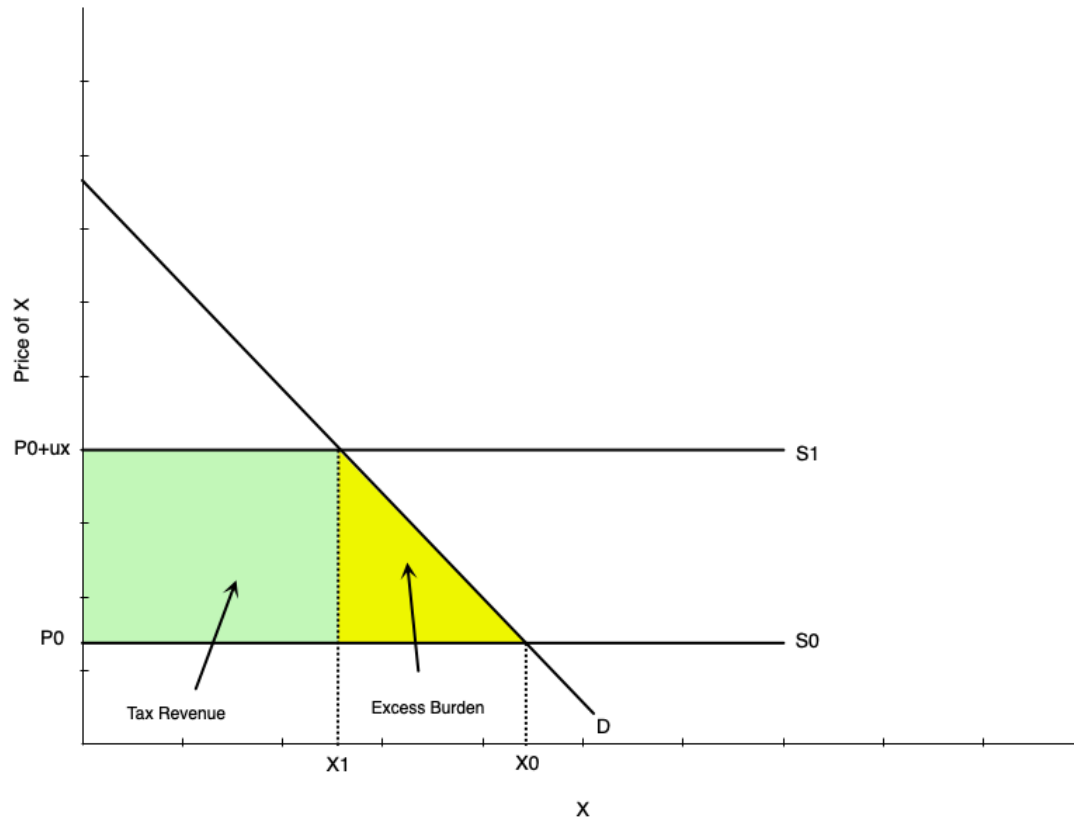
Tax goods so that the **percent reduction in quantity demanded is the same for all goods**. Equivalently, tax goods so that the marginal excess burden of the last dollar of revenue raised is the same for all goods.

- The next few slides establish this result

Ramsey Rule

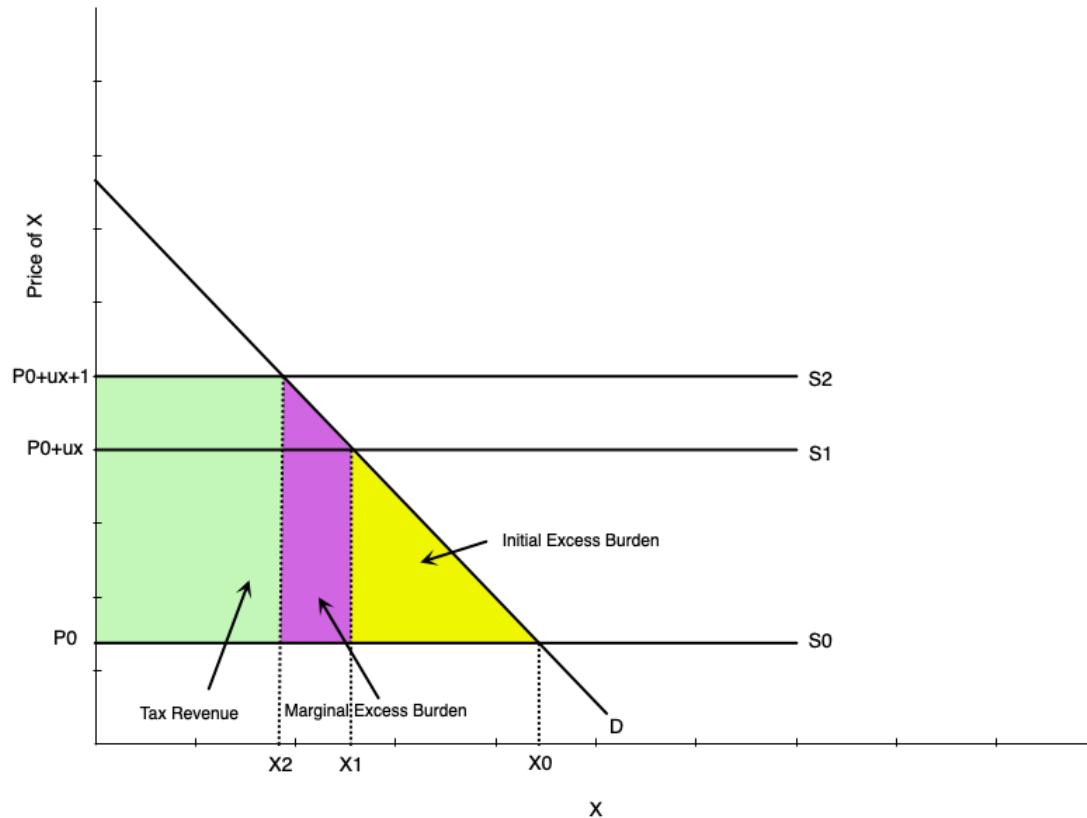
- Consider again two goods, X and Y , with prices P_X and P_Y
- Pretend that goods are neither substitutes nor complements
 - Simplifies the analysis because changes in their prices do not affect each other's demand
 - Results still hold for substitutes and complements, but analysis is more complex
- Also pretend the supply curve is horizontal
 - Consumers face the entire economic incidence of the tax
 - Again simplifies the analysis because we do not have to consider supply side effects
- Consider a unit tax u_X on good X

Ramsey Rule



- Consumer initially consumes X_0 at price P_0
- After tax u_X is imposed, price rises to $P_0 + u_X$
- Consumer reduces consumption to X_1
- As we have learned this involves an **excess burden**
 - Equals area of yellow triangle
 - Represents the loss in consumer surplus

Ramsey Rule



- Now imagine raising the tax further from u_X to $u_X + 1$
- The additional excess burden created is the **marginal excess burden**
 - Equals area of purple triangle
- Part of that excess burden was **tax revenue** (purple rectangle)
- Other part is additional **loss in consumer surplus** (purple triangle)

Ramsey Rule

- The **marginal excess burden** of the tax on good X is the purple area
 - Equals area of the purple triangle plus the purple rectangle
- To compute that mathematically, set $\Delta x = X_1 - X_2$
- The total area is then

$$\left(\frac{1}{2}\Delta x \times 1\right) + (\Delta x u_X)$$

- As noted in the textbook, if we pretend that $\frac{1}{2}\Delta x$ is very small, we can ignore it
 - So marginal excess burden is approximately $\Delta x u_X$

Ramsey Rule

- Going one step further, note that the slope of the demand curve is

$$\frac{u_X}{\Delta X} = \frac{1}{\Delta x}$$

- $\Delta X = X_0 - X_1$, the initial change in quantity demanded from the initial tax
- Both represent the rise over the run
- Rearranging gives

$$\Delta x u_X = \Delta X$$

- The marginal excess burden of the tax on good X is approximately ΔX
 - The change in quantity demanded from the initial tax

Ramsey Rule

- What about the change in revenue?
- Initially revenue was larger green rectangle plus purple rectangle
- After, it is larger green rectangle plus smaller green rectangle
- The **marginal tax revenue** is therefore the smaller green rectangle minus the purple rectangle

$$(X_2 \times 1) - (\Delta x u_X)$$

- We previously saw that $\Delta x u_X = \Delta X$
- We also know that $X_2 = X_1 - \Delta x$
- Subbing in gives us

$$X_1 - \Delta x - \Delta X$$

Ramsey Rule

- IF we pretend that Δx is very small, we can ignore it
 - So marginal tax revenue is approximately $X_1 - \Delta X$
- The marginal excess burden *per dollar of additional revenue* is

$$\frac{\Delta X}{X_1 - \Delta X}$$

- If we repeat this exact exercise for good Y , we find that the marginal excess burden per dollar of additional revenue is

$$\frac{\Delta Y}{Y_1 - \Delta Y}$$

Ramsey Rule

- If we set them equal

$$\frac{\Delta X}{X_1 - \Delta X} = \frac{\Delta Y}{Y_1 - \Delta Y}$$

- Which simplifies to the [Ramsey Rule](#)

$$\frac{\Delta X}{X_1} = \frac{\Delta Y}{Y_1}$$

- If you multiply both sides by 100 to express this as a percentage, it says percent change in quantity demanded of X equals percent change in quantity demanded of Y

Ramsey Rule with Elasticities

- Economists enjoy expressing things in terms of **elasticities**
- The price elasticity of demand is

$$\eta_X = \frac{\Delta X}{\Delta P_X} \times \frac{P_X}{X}$$

- Rearrange that to get

$$\frac{\Delta X}{X} = \eta_X \times \frac{\Delta P_X}{P_X}$$

Ramsey Rule with Elasticities

- In this case, consider instead an ad valorem tax that makes the price rise to $P_X(1 + t_X)$
- The numerator of this fraction is then $\Delta P_X = P_X(1 + t_X) - P_X = P_X t_X$
- Substituting into the equation on the previous slide gives

$$\frac{\Delta X}{X} = \eta_X \times t_X$$

Ramsey Rule with Elasticities

- The Ramsey Rule then becomes

$$\eta_X t_X = \eta_Y t_Y$$

- Equivalently

$$\frac{t_X}{t_Y} = \frac{\eta_Y}{\eta_X}$$

Ramsey Rule with Elasticities

! The Inverse Elasticity Rule

The optimal tax rate on a good is **inversely proportional** to its price elasticity of demand. If good Y has a high elasticity relative to good X , then t_X should be higher than t_Y .

- Why?
 - Because a tax on a good with a **high elasticity** will cause a large reduction in quantity demanded
 - This creates a **large excess burden**
 - So to minimize excess burden, tax goods with high elasticities **less**

Corlett-Hague Rule

- Recall that we could not tax leisure directly
 - And therefore could not apply an efficient (lump-sum equivalent) tax
 - This created the excess burden

The Corlett-Hague Rule

Approximate a leisure tax by taxing goods that are **complements** (used together with) to leisure. Examples: tax sporting equipment, household appliances, recreational vehicles, etc.

- This indirectly lowers demand for leisure and acts like a tax on leisure
- Gets us a bit closer to the **most efficient outcome**

Equity in Taxation

- Ramsey rule implies taxing goods that are **inelastic** more heavily
- Makes sense when those goods are socially undesirable (e.g., cigarettes, alcohol)
- But what if those goods are **necessities** (e.g., food, clothing, housing, medicine)?
- Problematic for a few reasons, but one is that it lacks **vertical equity**
 - **Vertical equity**: people with greater ability to pay should pay more in taxes
- You can modify the Ramsey rule to account for vertical equity
 - A modified rule may tax necessities less heavily
 - Essentially allows for **larger excess burden** to achieve **greater equity**

Summary

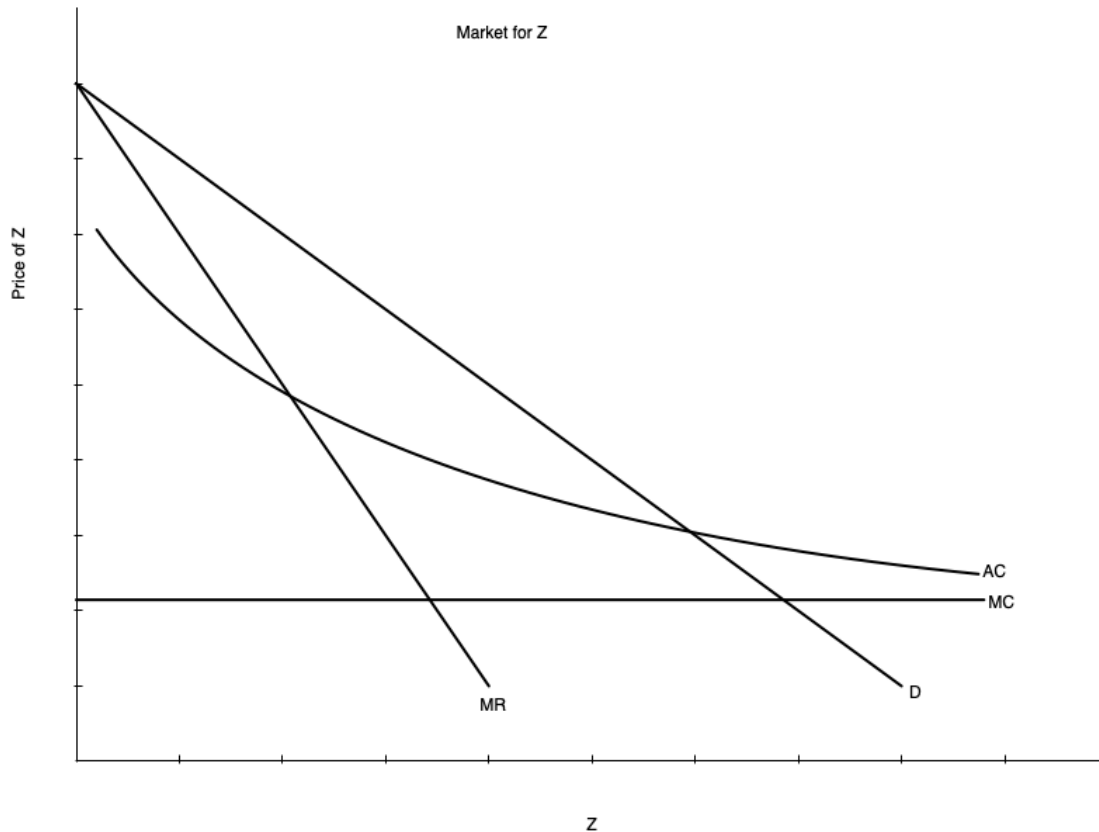
- In the real world we cannot achieve the efficient outcome of a **lump-sum tax** to raise government revenue
- Distortionary taxes change behaviour and create an **excess burden**
- We can design the tax system to minimize the excess burden
- The **Ramsey Rule** gives that optimal design
- But tax efficiency is not the sole consideration
 - **Equity** is also important
 - The optimal rule can change to account for equity considerations

Optimal User Fees

Introduction

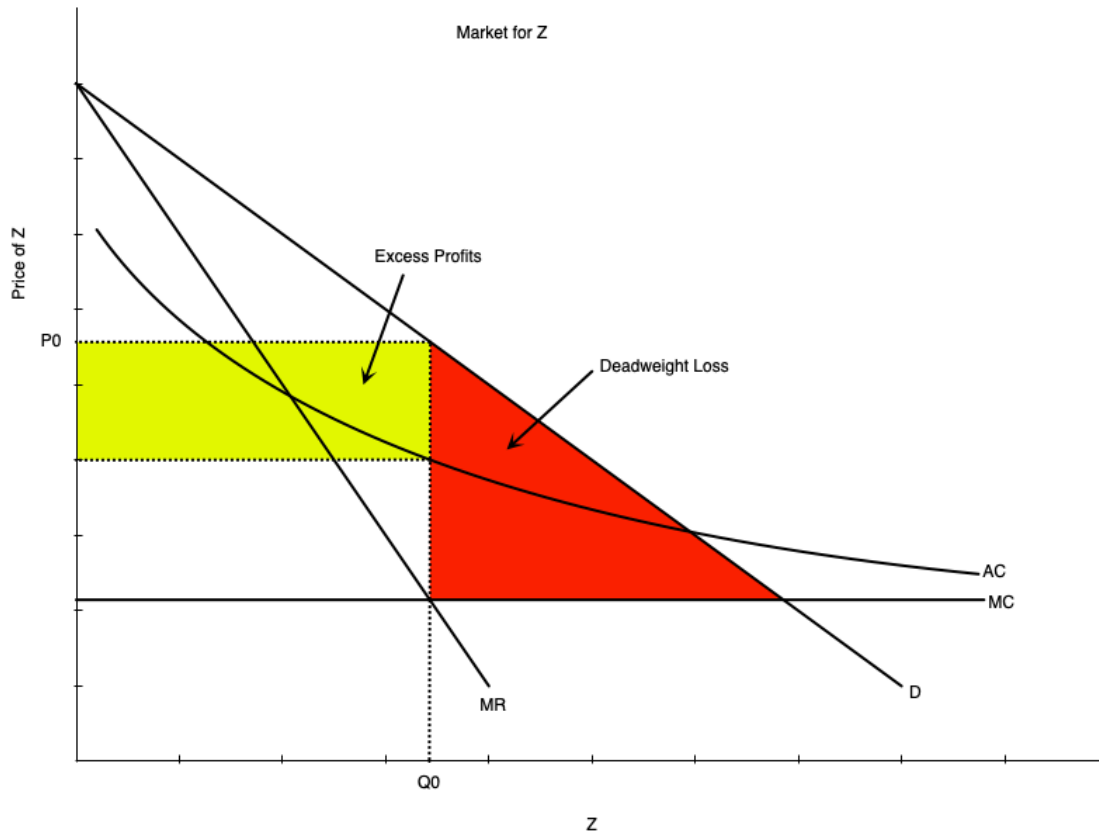
- Governments often charge **user fees** for public services
 - **User fee:** a fee charged by the government for the use of a good or service
 - Examples: park entrance fees, toll roads, public transit fares, etc.
- Like a tax, but not a tax
- Determining the optimal user fee is similar to determining the optimal tax

Government Production



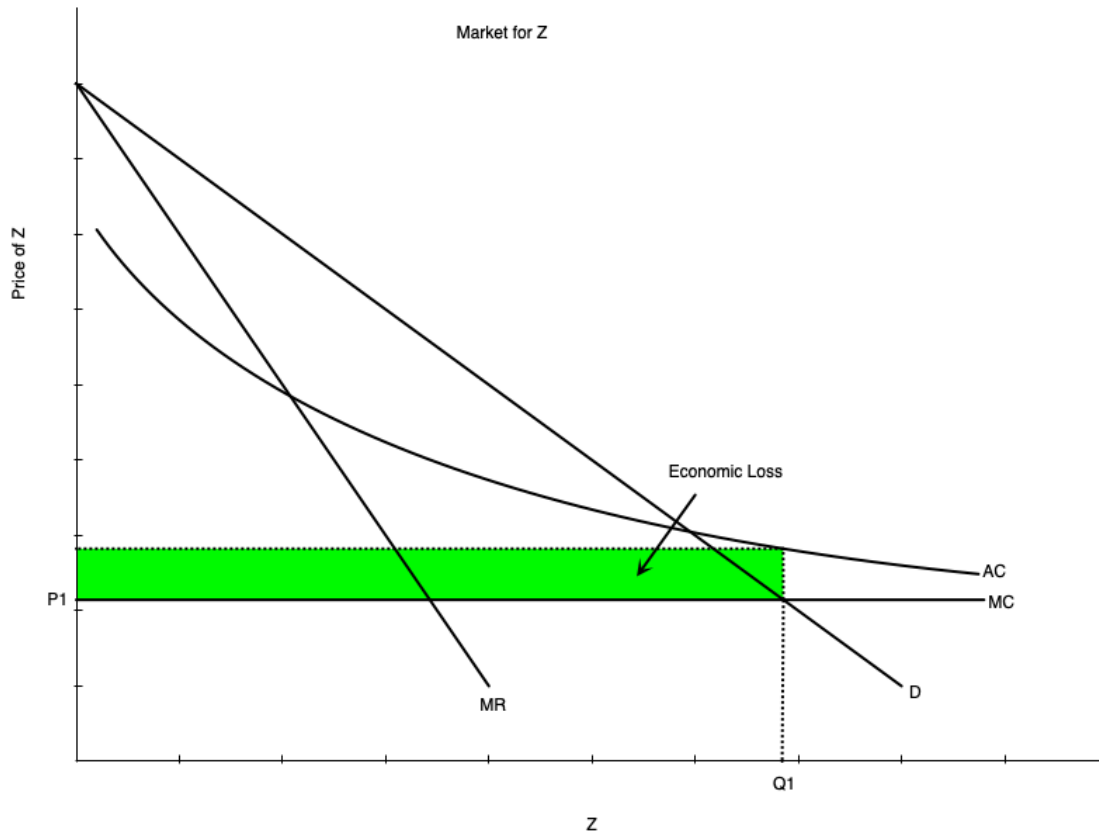
- Governments sometimes produce goods/services when there is a **natural monopoly**
 - **Natural Monopoly:** a market where a single firm can produce the entire output at lower cost than multiple firms
 - Happens with continuously decreasing average costs
 - Examples: water, electricity, public transit, etc.
- Natural monopoly depicted to the left
- Marginal cost fixed for ease of analysis
 - Could also draw it downward sloping as in the text

Government Production



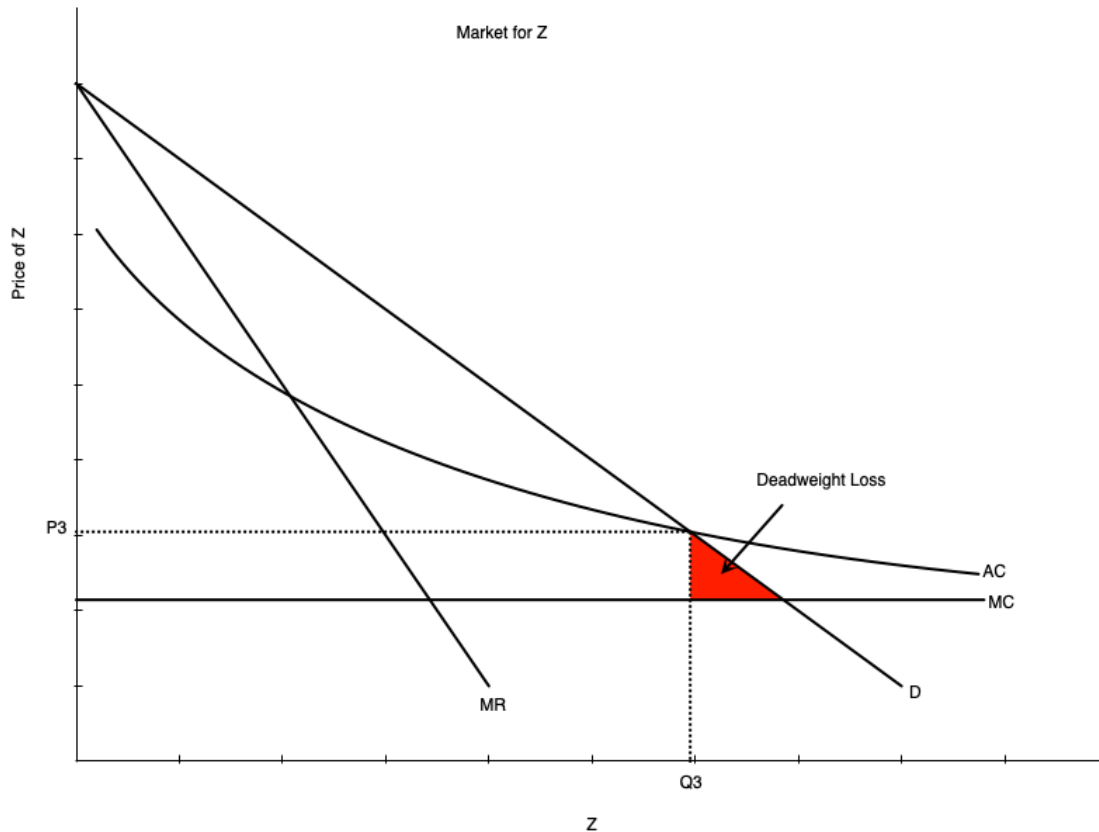
- An unregulated monopolist would produce Q_0 where $MR = MC$
- Would charge price P_0
- The monopoly earns **excess profits** equal to yellow area
- Also associated with **deadweight loss** (excess burden) equal to red area
- The monopolist does not produce the **efficient** ($P=MC$) level of Z

Government Production



- The government could take over and produce the good/service
 - Or regulate the private monopolist
- There are different options for production
- One is to produce the **efficient level** Q_1 where $P = MC$
- But price P_1 is below average cost AC
- The government would incur a **loss** equal to green area

Government Production



- Another option is **average cost pricing**
- Produce Q_2 where $P = AC$
- Government incurs **no excess profits or losses**
- Price P_2 is above MC , so there is still some **deadweight loss** (excess burden) equal to red area

Government Production

- A third option is to set $P=MC$ and charge a lump sum tax to cover the loss
 - Hard to do in practice because generally cannot levy lump sum taxes
 - Consumers not using the good would pay for it
- A fourth option is a **two-part tariff**
 - Set $P=MC$ and charge a fixed fee to cover the loss
 - Only users of the good pay the fee
 - Divide the fee by the number of users to get the fee per user

Government Production

- Lastly, use the [Ramsey Rule](#)
 - If government produces multiple goods/services, set user fees for each
 - Optimally set them so that the percent reduction in quantity demanded is the same for all goods/services
 - We saw that this involves setting higher fees for goods/services with [inelastic demand](#)
 - The user fee effectively acts like a tax on the good/service

Optimal Income Tax

Introduction

- So far we have discussed optimal commodity taxes and user fees
- **Income taxes** are another major source of government revenue
- Income taxes are different because they tax a person's **ability to pay**
 - Ability to pay is determined by income
 - Income taxes therefore affect both **labour supply** and **consumption**
- You can design an income tax optimally

Old School Optimal Tax

- A very old approach to optimal income taxation is [Edgeworth's Model](#)

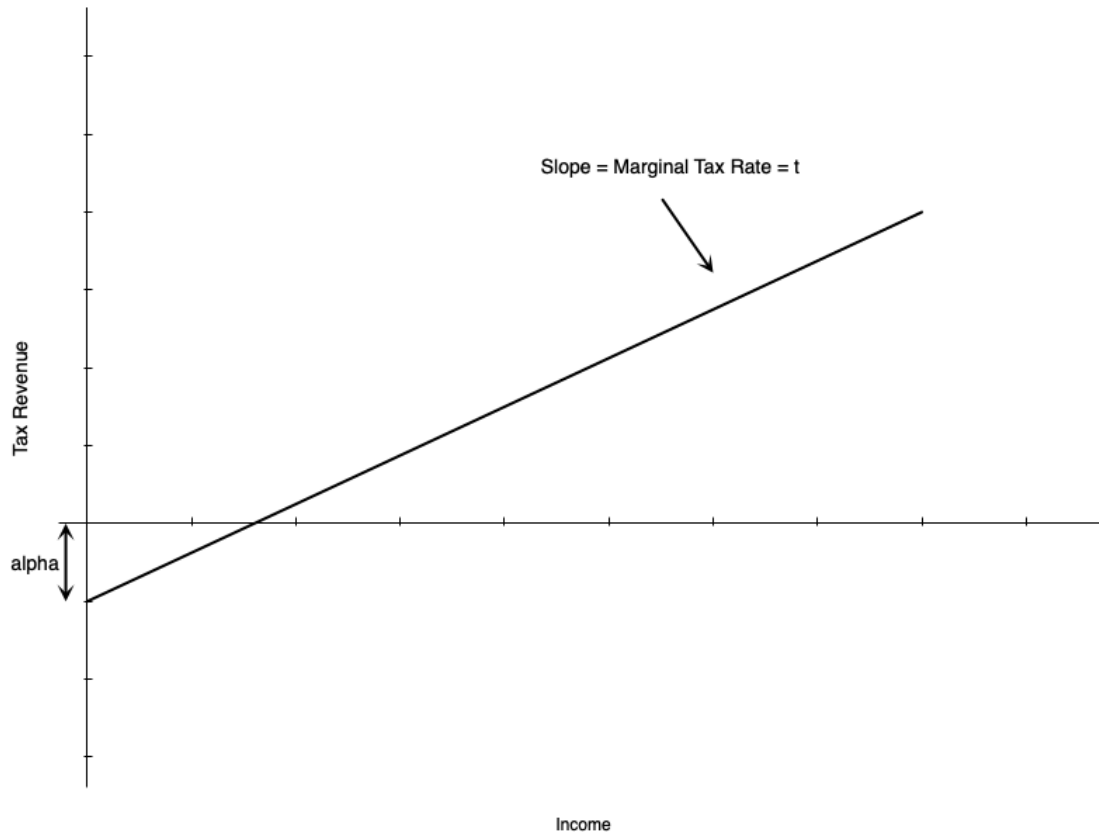
Edgeworth's Model Assumptions

- Maximize [social welfare](#): the combination of individual utilities
 - Social welfare is [utilitarian](#): the sum of individual utilities
 - Each person's utility is the same, depends only on income
 - Has [diminishing marginal utility](#) of income
 - Total amount of income in society is fixed
-
- This model predicts that after-tax income is the same for everyone
 - Involves taxing rich and giving to poor until equality is reached
 - This is [not a realistic model](#), but it is a starting point

Newer Approach to Optimal Tax

- Major problem with Edgeworth's model is that it **ignores behavioural responses**
 - People will supply less labour if taxed heavily
- We know from previous sections that taxes affect labour supply
- Modern models take this into account
 - Taxes raise **revenue**
 - But also create **excess burden** by distorting labour supply
 - Optimal tax balances these two effects
- These models compute optimal tax in a world where people choose labour and leisure

Newer Approach to Optimal Tax

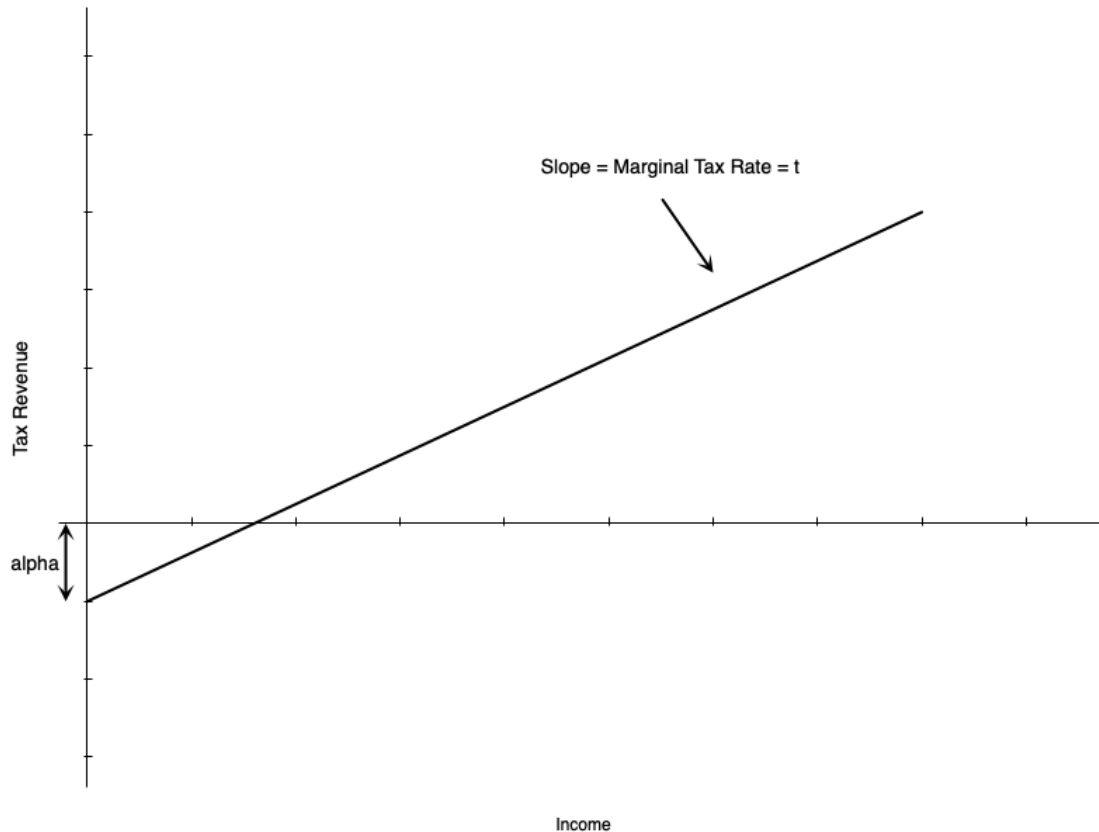


- A simplified version of this model is depicted to the left
- In this setup revenue is

$$Revenue = -\alpha + t \times Income$$

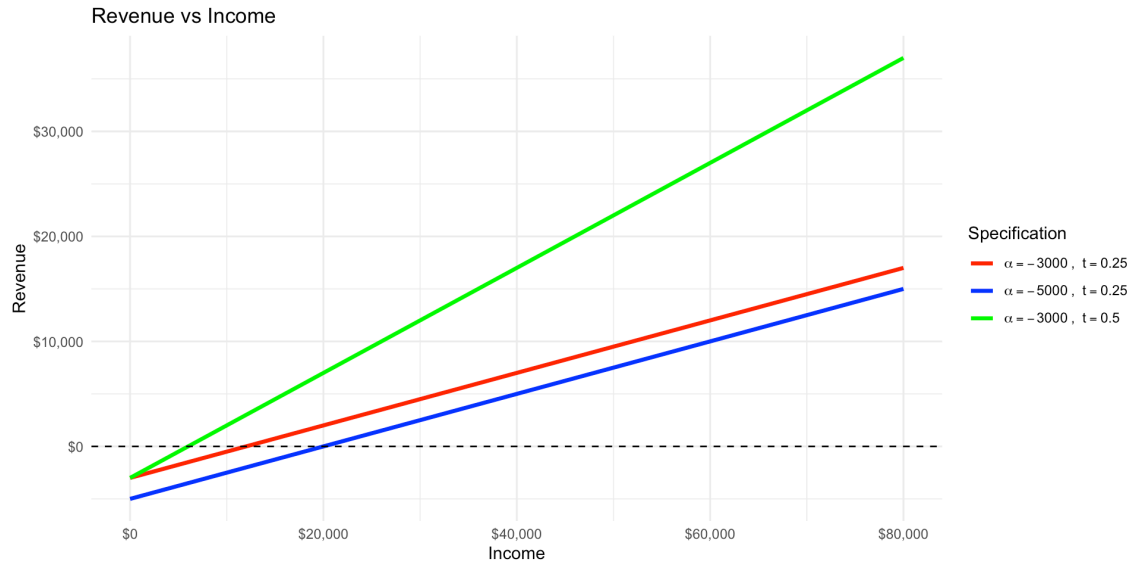
- The term $-\alpha$ is a fixed payment to everyone
 - If $\alpha > 0$, it is a **subsidy**
 - If $\alpha < 0$, it is a **lump-sum tax**
- The term t is the marginal income tax

Newer Approach to Optimal Tax



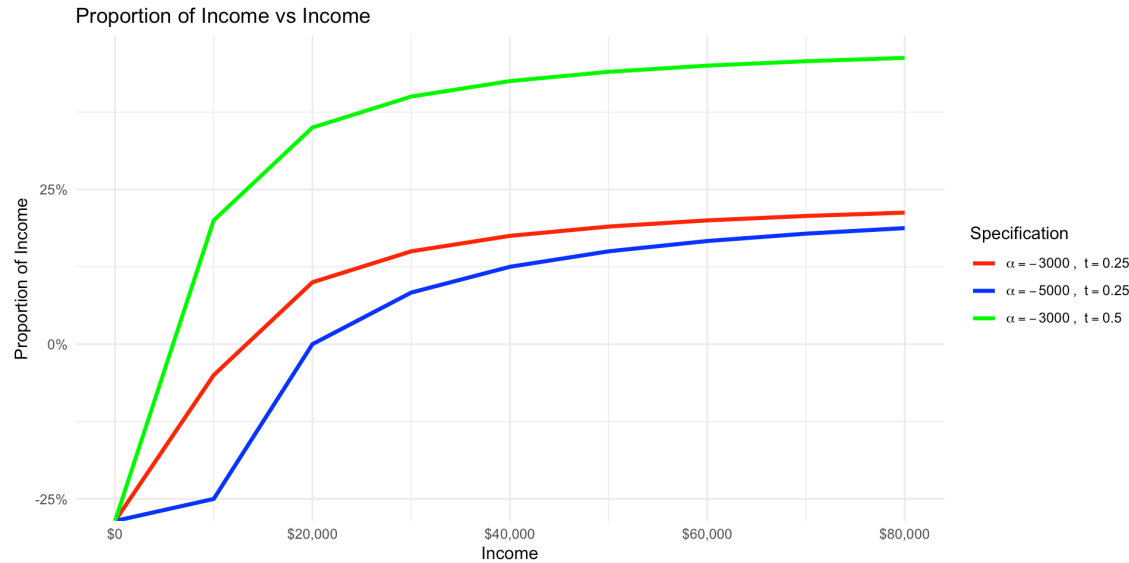
- A constant marginal tax is a **flat tax**
 - Comes from a linear tax schedule
- Studies find that the optimal income tax is **non-linear**
 - Marginal tax rate rises with income
 - Implies a **progressive tax system**
 - This exists in many countries, including Canada and the US
- However, they also find that the optimal non-linear tax is approximated a flat tax
 - So we can still gain insights from the flat tax

Newer Approach to Optimal Tax



- An example with different flat taxes is shown to the left
- Graph plots total revenue
- There are three curves
 - $\alpha = -3000, t = 0.25$
 - $\alpha = -5000, t = 0.25$
 - $\alpha = -3000, t = 0.5$
- α changes the intercept
- t changes the slope

Newer Approach to Optimal Tax



- Now plot the proportion of income paid in tax
- Here you can see the **progressivity** of the tax
- The tax with the highest marginal tax rate ($t = 0.5$) and the lowest fixed payment ($\alpha = -3000$) is the most progressive
- The higher the marginal tax rate, the **more progressive** the tax
- The lower the fixed payment, the **more progressive** the tax

Newer Approach to Optimal Tax

- The specific optimal tax rates depend on the model assumptions
 - Labour supply elasticity
 - Social welfare function
 - Ability to pay
 - Income distribution
- With the utilitarian social welfare function, optimal is around 50%
 - Optimal grant α is about 60% of average worker income
- Using a more egalitarian social welfare function, optimal tax is higher
 - Egalitarian social welfare function puts more weight on poorer individuals
 - Optimal rate around 80%
- Finally, some studies find that if tax rates are allowed to be different, rich should pay lower marginal rate

Politics and Time Inconsistency

Introduction

- **Optimal tax theory** provides purely theoretical guidance on how to design taxes
- In practice, tax policy is determined by **politics**
 - Politicians care about getting elected
 - Voters care about their own taxes and benefits
- So the optimal tax is **not a realistic outcome**
- It is possible that in the real world, implementing the optimal tax is not even desirable

Time Inconsistency Problem

- Imagine a government wants to tax society to raise revenue
- There are three possible goods that people can consume
 - X , Y , and Leisure L
- Labour has a fixed supply so income is also fixed
- The government currently taxes only X , not Y
- A tax economist suggests lowering the tax on X and taxing both X and Y at the same rate
 - **Efficient** because relative prices are unchanged, no change in labour
 - No excess burden

Time Inconsistency Problem

- Citizens are cynical
 - They think the government will implement the tax on Y but **not lower it on X**
 - View the government as trying to maximize revenue at their expense
 - In certain situations they might be right
- If they are correct, taxing only X is more efficient than the alternative of taxing both

Time Inconsistency Problem

Optimal tax policy is **not credible** over time. The government cannot commit to future policies. To fully implement optimal tax, the government must credibly commit to future policies – this is difficult in practice, so optimal tax may not be achievable or even desirable.

Other Criteria for Tax Design

Horizontal Equity

! Horizontal Equity

People in “equal positions” should be treated equally. When it comes to taxation, equal positions might mean equal ability to pay, out of income, consumption, or wealth.

- Horizontal equity is desirable because it is viewed as **fair**
 - People in equal positions should pay the same tax
 - If not, people may view the tax system as **unfair** and try to avoid taxes
- Defining horizontal equity in terms of income or wage is **problematic**
 - Two people who earn the same wage but work different hours have different abilities to pay
 - Taxing wage instead is a problem because it is based on human capital investments

Horizontal Equity

- Could instead define horizontal equity in terms of **utility**
 - **Utility definition of horizontal equity:** people with the same pre-tax utility should have the same after-tax utility, and taxes should not alter the utility ordering
- Several problems with this approach
 - Utility is **unobservable**
 - If taxing only income, it can penalize people who
 - Consume goods that are income intensive
 - Work in jobs where the pay is purely monetary instead of non-income benefits (e.g. nice work environment, flexible work)
 - Depends on utility ordering prior to any new taxes being imposed (i.e. biased towards status quo)
- Basic idea is that while horizontal equity is desirable, it is **hard to define and implement** in practice



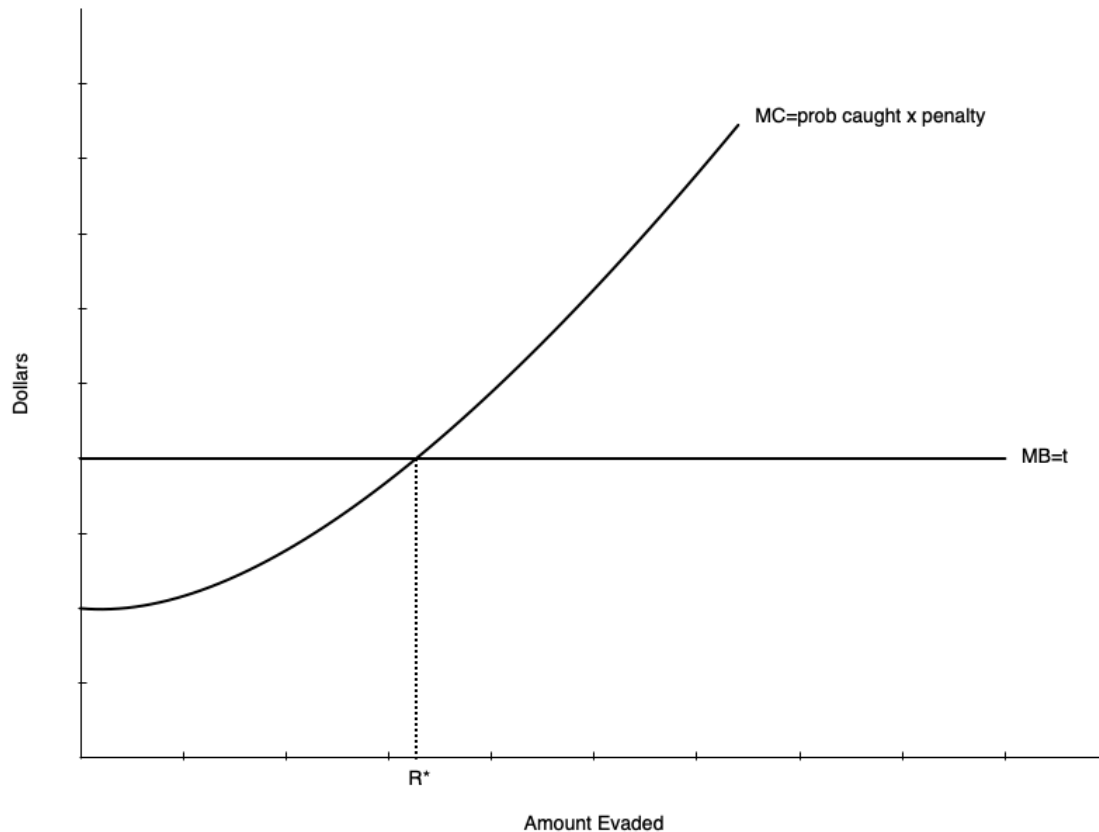
Administrative Costs

- Taxes are **costly to administer**
 - Direct costs of about \$1 to collect \$100 of tax revenue in Canada
 - Average taxpayer pays about \$200-\$250 in tax preparation costs per year
 - Overall it costs about **2% of GDP** to collect taxes in Canada
- Tax design must balance revenue benefits against administrative costs
- Some changes have led to **smaller costs**
 - Online filing, automatic download of tax forms
 - Potential for future: automatic filing

Compliance

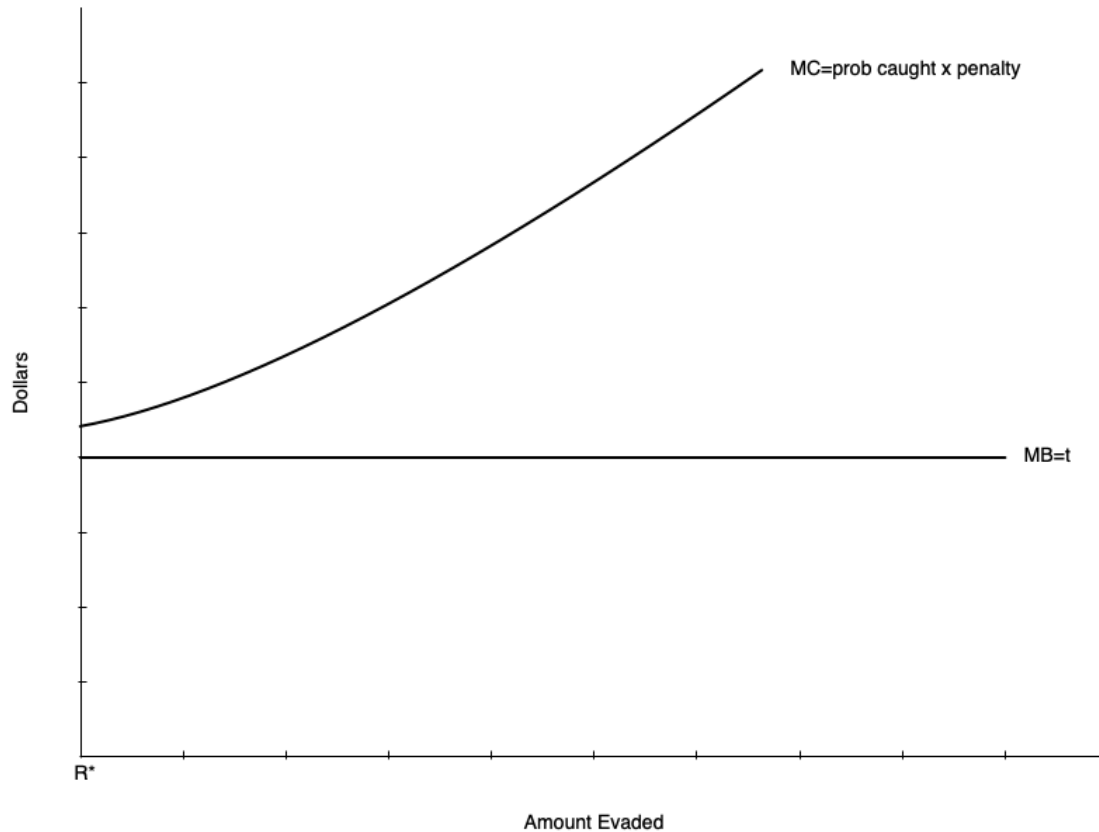
- Canada and other countries operate on a [self-assessment system](#)
 - Taxpayers are responsible for reporting their income and calculating their taxes
 - Government audits a small percentage of taxpayers to ensure compliance
- Some people engage in **tax avoidance**
 - **Legal ways**: changing behaviour to minimize taxes
 - **Illegal ways**: tax evasion, not reporting income, fraud
- It is not always easy to catch tax evasion
 - Government has to balance the costs of enforcement against the benefits
 - If enforcement is too costly, it may not be worth it
- There is an economic theory that determines “optimal” tax evasion

Compliance



- Imagine that costs of evasion rise with the amount evaded
 - Penalty rises with amount evaded
 - There is some probability that you get caught
 - Could be psychic costs to cheating
 - Risk aversion might play a role
- Benefit is constant
 - Gain is t per \$1 evaded
- As with many economic decisions, optimal evasion occurs where **marginal benefit equals marginal cost**
- This person would evade R^*

Compliance



- If costs are too high, **no evasion** occurs
 - Probability of getting caught might be high
 - Penalty might be high
 - Large psychic costs or very risk averse

Compliance

- Also the normative question about tax evasion: do we care?
- Some economies have large **underground or informal economies**
 - Developing countries often have significant employment in informal economies
- While it is incorrect to call them tax evaders, people in informal employment do not pay taxes
- It might be desirable to have the underground/informal economy even if they pay no tax
 - They are often **poor** and would not pay much tax
 - They often provide goods/services that are **beneficial to society**
 - Trying to eliminate the informal economy might do **more harm than good**

Summary

Summary

- Taxes create **excess burden** by distorting behaviour
- The **Ramsey Rule** gives the optimal way to tax goods to minimize excess burden
- **User fees** can be optimized in a similar manner
- Optimal income tax is **progressive**, but the specific rates depend on model assumptions
- **Politics** and **time inconsistency** can prevent optimal tax from being implemented
- Other criteria for tax design include **horizontal equity**, **administrative costs**, and **compliance**

References

References

- Rosen, Harvey S., and Lindsay M. Tedds, and Trevor Tombe, and Jean-Francois Wen, and Tracy Snoddon. Public Finance in Canada. 6th Canadian edition. McGraw-Hill Ryerson, 2023.
- Gruber, Jonathan. Public Finance and Public Policy. 7th edition. Worth Publishers, 2022.